

2(a)(i)	prokaryotic cells are (usually) 1–5 μm (in length) (A any value within the 1–5 range) or prokaryotic cells are smaller than, <i>P. myojinensis</i> / the cell in Fig. 2.1 ; AVP ; e.g. <i>suggestion that</i> (typical prokaryotic) cell contains plasmids qualified <i>ref. to</i> periplasmic space	1
2(a)(ii)	I endosymbionts <i>any two from:</i> ribosomes are, present in the nucleus of <i>P. myojinensis</i> / not present in the nucleus of a eukaryotic cell ; <i>P. myojinensis</i> has only one nuclear membrane v a eukaryotic cell has a nuclear envelope ; A double membrane / two nuclear membranes, for envelope A does not have double membrane-bound organelles named organelle, not present in <i>P. myojinensis</i> / present in a eukaryotic cell ; e.g. mitochondrion (rough / smooth) endoplasmic reticulum Golgi body vesicles / vacuoles nucleolus chloroplast, <i>if qualified</i> <i>P. myojinensis</i> has a cell wall and some eukaryotic cells do not have a cell wall ;	2

Question	Answer	Marks
2(a)(iii)	any two from: <i>context is cell wall composition</i> peptidoglycan / murein, so, similar to, bacteria / prokaryotes different to, eukaryotes (with cell walls) / plants / fungi ; cellulose / chitin, so, similarity to eukaryotes bod plants / fungi difference to, bacteria / prokaryotes ; AVP ; e.g. mannan and glucan similarity to eukaryotes <i>ref. to composition of Gram negative bacterial cell walls</i> no, peptidoglycan / murein, could be similar to, Archaea / archaeans, which are prokaryotes	2
2(b)	use (stage) micrometer to, work out / estimate / AW, length / number of micrometres, per, eyepiece / graticule, division ; A calibrate the eyepiece graticule with a stage micrometer convert number of (eyepiece graticule) divisions (along cell length) to micrometre measurement / AW ;	2